

# ALESSANDRO CASTELLANI

Milan / Como, Italy · +39 333 528 8084 · alecaste041202@gmail.com

Digital Hub: apmcastellani-projects.github.io · LinkedIn: in/alessandro-castellani · GitHub: apmcastellani-projects

## EDUCATION

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| <b>Università Cattolica del Sacro Cuore</b><br><i>Master of Science (Laurea Magistrale) in Data Analytics for Business</i><br>Coursework: Applied Linear Models, Algorithms & Deep Learning, Time Series, Computational Statistics, Optimization, Statistical Learning.                                                                            | <i>Milan, Italy</i><br><i>Sep 2025 – Expected 2027</i> |
| <b>Università degli Studi dell’Insubria</b><br><i>Bachelor of Science (Laurea Triennale) in Mathematics</i><br><b>Thesis:</b> <i>Neural Network Architectures for Football Performance Forecasting using TensorFlow and Keras.</i> Modeled historical player metrics to forecast seasonal performance using backpropagation and Adam optimization. | <i>Como, Italy</i><br><i>Graduated – Mar 2025</i>      |
| <b>Pontificio Collegio Gallio</b><br><i>Liceo Scientifico Internazionale (High School Diploma)</i>                                                                                                                                                                                                                                                 | <i>Como, Italy</i><br><i>Sep 2016 – Jun 2020</i>       |

## FEATURED PROJECTS & DATA COMPETITIONS

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| <b>Evaluating Offensive Production in Serie A (xG Linear Modeling)</b><br><i>Technologies: R, FBref/Opta Event Data, Multiple OLS Regression, Residual Diagnostics</i>                                                                                                                                                                                                                                                                                                                          | <i>Academic Research (Cattolica)</i><br><i>2026</i>   |
| <ul style="list-style-type: none"><li>Developed a log-linear model on 120 Serie A forwards analyzing Expected Goals (xG) generation beyond raw shot volume (<math>R^2 = 91.4\%</math>, Adjusted <math>R^2 = 90.8\%</math>).</li><li>Conducted tactical outlier diagnostics (identifying tactical roles of Rade Krunic, João Pedro, Patrick Cutrone), restoring residual normality (<math>p = 0.0557</math>), and validated model stability out-of-sample on Olivier Giroud (2022/23).</li></ul> |                                                       |
| <b>IFCS 2026 International Data Challenge</b><br><i>Technologies: Python, Scikit-Learn, Pandas, Imbalanced Classification</i>                                                                                                                                                                                                                                                                                                                                                                   | <i>Data Science Competition</i><br><i>2026</i>        |
| <ul style="list-style-type: none"><li>Engineered a machine learning classification pipeline predicting financial distress across 14,000 Italian SMEs.</li><li>Formulated signed logarithmic transformations (<math>\text{sign}(x) \cdot \ln(1 +  x )</math>) for heavy-tailed balance sheets, 5-Fold Stratified Cross-Validation without leakage, and optimized decision thresholds via Youden’s J Index and ROC metrics.</li></ul>                                                             |                                                       |
| <b>Multi-Horizon XGBoost Predictive Engine &amp; Dashboard</b><br><i>Technologies: Python, FastAPI, XGBoost, Purged Walk-Forward Cross-Validation, Plotly</i>                                                                                                                                                                                                                                                                                                                                   | <i>Independent ML Engineering</i><br><i>2026</i>      |
| <ul style="list-style-type: none"><li>Built an end-to-end time-series return predictor utilizing purged walk-forward cross-validation with embargo buffers to eliminate temporal lookahead bias, deployed with a FastAPI REST backend and interactive web dashboard.</li></ul>                                                                                                                                                                                                                  |                                                       |
| <b>Serie A Player Profiling &amp; Longitudinal Analytics</b><br><i>Technologies: Python, Pandas, Matplotlib, Plotly</i>                                                                                                                                                                                                                                                                                                                                                                         | <i>Football Analytics Suite</i><br><i>2024 – 2026</i> |
| <ul style="list-style-type: none"><li>Analyzed 5 seasons of defender performance metrics (2019–2023) and KPI correlation structures for attacking and goalkeeping roles.</li></ul>                                                                                                                                                                                                                                                                                                              |                                                       |

## WORK EXPERIENCE

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| <b>XROAD SSDRL</b><br><i>Collaboratore Amministrativo (Part-Time)</i>                                                                                         | <i>Cadorago, Italy</i><br><i>Jan 2025 – Present</i>     |
| <ul style="list-style-type: none"><li>Front-office administration, customer interaction, and active subscription coordination for over 200 members.</li></ul> |                                                         |
| <b>Private Mathematics Tutoring</b><br><i>Mathematics Tutor</i>                                                                                               | <i>Milan / Como, Italy</i><br><i>Sep 2020 – Present</i> |
| <ul style="list-style-type: none"><li>Mentored 50+ students across secondary and university levels in mathematics, calculus, and linear algebra.</li></ul>    |                                                         |

## ATHLETICS & EXTRACURRICULAR

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| <ul style="list-style-type: none"><li><b>Competitive Golf Athlete:</b> Active player since 2014. Represented Lombardy at the Italian National Regional Championships and competed internationally for Italy. Cultivated deep concentration, strategic planning, and composure under pressure.</li><li><b>Hyrox Competitor:</b> Functional fitness and endurance competitor (8km running + 8 functional workout stations).</li></ul> |  |
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## TECHNICAL SKILLS & COMPETENCIES

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| <ul style="list-style-type: none"><li><b>Programming &amp; Libraries:</b> Python (NumPy, Pandas, Scikit-Learn, TensorFlow, Keras, FastAPI), R (tidyverse, car, leaps), SQL, Git, L<sup>A</sup>T<sub>E</sub>X.</li><li><b>Domain Expertise:</b> Football Analytics (Opta/FBref event metrics, xG/xA modeling, tactical profiling), Financial Modeling, Time-Series.</li><li><b>Languages:</b> Italian (Native), English (Fluent / Professional Working Proficiency).</li></ul> |  |
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